

C950

Data Structures and Algorithms II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A delivery firm serving 12 towns models roads as a weighted graph: A-B 4, A-C 1, C-B 2, B-D 1, C-D 6 (miles). Trucks leave at 6 a.m. What is the shortest distance from A to D?

- A. 5 miles
- B. 7 miles
- C. 4 miles
- D. 6 miles

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A graph includes an edge with weight -3 and no negative cycles. Which method still finds correct shortest paths?

- A. Breadth-first search
- B. Bellman-Ford
- C. Dijkstra's algorithm, which always finds the shortest path
- D. Prim's algorithm

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A professional networking site launched in 2019 now has 3 million members. It wants to show how many introductions separate any two members, where every connection counts the same. Which algorithm finds the fewest connections?

- A. A minimum spanning tree algorithm
- B. Any traversal, since all traversals visit every node
- C. Breadth-first search
- D. Depth-first search

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A recursive $\text{fib}(n)$ written in Python 3.12 on a laptop with 16 GB of RAM calls $\text{fib}(n - 1) + \text{fib}(n - 2)$ and is very slow for $n = 40$. A developer adds a dictionary that stores each result the first time it is computed. What is the new time complexity?

- A. $O(n)$
- B. $O(1)$
- C. $O(2^n)$
- D. $O(\log n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Merge sort splits a list into halves that never share any work. Why does adding memoization not speed it up?

- A. Memoization only ever works on numeric results.
- B. Its subproblems do not overlap, so nothing is reused.
- C. Python's recursion limit blocks caching.
- D. It lacks optimal substructure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A courier company whose drivers average nine years of experience must route one truck through 40 stops and return to the depot using the shortest possible total distance. Which statement is correct?

- A. Dijkstra's algorithm solves it exactly.
- B. It has been proven that no polynomial algorithm can ever exist.
- C. No known algorithm solves it exactly in polynomial time, so a heuristic such as nearest neighbor is used.
- D. Brute force works well because 40 stops is small.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A log file from a server farm in Nevada contains 100 symbols: A appears 50 times, B 25, C 15, and D 10. The file is backed up nightly. Using Huffman coding, how many bits encode the file?

- A. 175
- B. 100
- C. 200
- D. 265

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

In a Huffman code, symbol X has the shortest code of all. What does this tell you about X?

- A. X makes up more than half of the text.
- B. X comes first alphabetically.
- C. X is the least frequent symbol.
- D. X is among the most frequent symbols.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A package tracker loads 40 packages from a CSV into a chaining hash table before the truck leaves at 8:00 a.m. A bug makes the hash function return 0 for every package ID. What is the worst-case lookup time?

- A. $O(n^2)$
- B. $O(\log n)$
- C. $O(1)$, because hash tables are always constant time
- D. $O(n)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A news site serving 5 million readers caches articles in a splay tree, because readers often reopen stories they just read. After the tree accesses the article with key 17, where is 17?

- A. At a leaf
- B. Wherever keeps subtree heights within one
- C. In the same place as before
- D. At the root

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A campus is linking 6 buildings with fiber using Kruskal's algorithm and union-find. The contractor offers a discount on runs over 500 feet. The next cheapest edge, \$3,000, joins buildings B and E, which are already in the same set. What should the algorithm do?

- A.** Skip the edge, because it would form a cycle.
- B.** Stop, because the tree must be complete.
- C.** Add it and remove the most expensive edge in the cycle.
- D.** Add it, since it is the cheapest remaining edge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A minimum spanning tree has 11 edges. How many vertices does the graph have?

- A.** 22
- B.** 12
- C.** 11
- D.** 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A delivery firm ~~serving 12 towns~~ models roads as a weighted graph: A-B 4, A-C 1, C-B 2, B-D 1, C-D 6 (miles). ~~Trucks leave at 6 a.m.~~ What is the shortest distance from A to D?

Target word: shortest path weight A to D **Decides:** A-C-B-D = 1 + 2 + 1 = 4.

A	Confusion set	A→B→D uses few hops but weighs 5.
B	Dropped element	A→C→D skips the cheaper C-B-D detour.
C	Correct	A→C (1), C→B (2), B→D (1) totals 4.
D	Wrong stem number	The C-D edge weight alone.

2. Answer B Concept 1 · transfer

A graph includes an edge with weight -3 and no negative cycles. Which method still finds correct shortest paths?

Target word: algorithm for negative weights **Decides:** Dijkstra assumes non-negative weights.

A	Adjacent category	Only counts edges; ignores weights.
B	Correct	Relaxes all edges repeatedly and handles negative weights.
C	Absolute language	Can fail with negative edges.
D	Confusion set	Builds a minimum spanning tree, not shortest paths.

3. Answer C Concept 2

A professional networking site ~~launched in 2019 now has 3 million members~~. It wants to show how many introductions separate any two members, where every connection counts the same. Which algorithm finds the fewest connections?

Target word: fewest edges, unweighted **Decides:** Unweighted graph, minimum hops.

A	Adjacent category	Minimizes total tree weight, not the path between two members.
B	Absolute language	Visiting every node does not mean finding the shortest path.
C	Correct	BFS explores level by level, so the first time it reaches a node is by the fewest edges.
D	Confusion set	DFS may find a long path first.

4. Answer A Concept 3

A recursive `fib(n)` ~~written in Python 3.12 on a laptop with 16 GB of RAM~~ calls `fib(n - 1) + fib(n - 2)` and is very slow for $n = 40$. A developer adds a dictionary that stores each result the first time it is computed. What is the new time complexity?

Target word: complexity after memoization **Decides:** Each of the n subproblems is solved once.

A	Correct	Each <code>fib(k)</code> for $k \leq n$ is computed once and reused.
B	Confusion set	Confuses one cached lookup with filling all n entries.
C	Dropped element	Ignores the cache, which removes the repeated calls.
D	Adjacent category	Possible with matrix methods, not with simple memoization.

5. Answer B Concept 3 • transfer

Merge sort splits a list into halves that never share any work. Why does adding memoization not speed it up?

Target word: why DP gives no gain **Decides:** No overlapping subproblems.

A	Absolute language	Any hashable subproblem key can be cached.
B	Correct	Memoization helps only when the same subproblem repeats.
C	Adjacent category	Recursion depth is unrelated to whether caching helps.
D	Confusion set	Merge sort combines optimal sub-results; overlap is what is missing.

6. Answer C Concept 4

A courier company ~~whose drivers average nine years of experience~~ must route one truck through 40 stops and return to the depot using the shortest possible total distance. Which statement is correct?

Target word: nature of the routing problem **Decides:** It is the traveling salesperson problem, which is NP-hard.

A	Confusion set	Dijkstra finds point-to-point paths, not a tour of all stops.
B	Absolute language	Whether $P = NP$ is still unresolved.
C	Correct	TSP is NP-hard; heuristics give good, fast routes.
D	Dropped element	Ignores factorial growth: $39!$ routes is astronomically large.

7. Answer A Concept 5

A log file ~~from a server farm in Nevada~~ contains 100 symbols: A appears 50 times, B 25, C 15, and D 10. ~~The file is backed up nightly.~~ Using Huffman coding, how many bits encode the file?

Target word: total Huffman bits **Decides:** Codes: $A = 1$ bit, $B = 2$, $C = 3$, $D = 3$.

A	Correct	$50 \times 1 + 25 \times 2 + 15 \times 3 + 10 \times 3 = 175$.
B	Wrong stem number	The symbol count, not bits.
C	Adjacent category	Fixed-length 2-bit codes: 100×2 .
D	Inverted operation	Gives the shortest codes to the rarest symbols.

8. Answer D Concept 5 • transfer

In a Huffman code, symbol X has the shortest code of all. What does this tell you about X?

Target word: infer frequency from code length **Decides:** Shortest codes go to the most frequent symbols.

A	Absolute language	Being most frequent does not require a majority.
B	Confusion set	Huffman uses frequency, not symbol order.
C	Inverted operation	Rare symbols get the longest codes.
D	Correct	Huffman places frequent symbols near the root.

9. Answer D Concept 6

A package tracker loads 40 packages from a CSV into a chaining hash table before the truck leaves at 8:00 a.m. A bug makes the hash function return 0 for every package ID. What is the worst-case lookup time?

Target word: lookup cost with total collision **Decides:** Every key sits in one chain.

A	Confusion set	That is the cost of n lookups, not one.
B	Adjacent category	That is a balanced tree lookup.
C	Absolute language	O(1) is the average with a good hash, not a guarantee.
D	Correct	Lookup must scan a single chain of n items.

10. Answer D Concept 7

A news site serving 5 million readers caches articles in a splay tree, because readers often reopen stories they just read. After the tree accesses the article with key 17, where is 17?

Target word: position after splay **Decides:** Splaying moves the accessed node to the root.

A	Inverted operation	Splaying moves nodes up, not down.
B	Adjacent category	That is the AVL balance rule.
C	Confusion set	That is a plain BST.
D	Correct	Every access splays the node to the root.

11. Answer A Concept 8

A campus is linking 6 buildings with fiber using Kruskal's algorithm and union-find. The contractor offers a discount on runs over 500 feet. The next cheapest edge, \$3,000, joins buildings B and E, which are already in the same set. What should the algorithm do?

Target word: action on a cycle-forming edge **Decides:** B and E are already connected.

A	Correct	Kruskal only joins different sets.
B	Confusion set	The tree is complete only at $V - 1$ edges.
C	Adjacent category	That is a different MST repair approach, not Kruskal.
D	Dropped element	Leaves out the cycle check.

12. Answer B Concept 8 · transfer

A minimum spanning tree has 11 edges. How many vertices does the graph have?

Target word: solve for V **Decides:** $\text{Edges} = V - 1$.

A	Confusion set	Counts edge endpoints.
B	Correct	$11 + 1 = 12$.
C	Wrong stem number	The edge count.
D	Inverted operation	Subtracts one instead of adding.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Dijkstra's shortest path	Dijkstra finds the lowest total-weight path from a source when all edge weights are non-negative.	Picking the route with the fewest hops or cheapest first edge.	A weighted graph and a "shortest" or "cheapest" route.
2	BFS vs. DFS	Breadth-first search (queue) finds the fewest-edge path in an unweighted graph; depth-first search (stack) does not guarantee it.	Using DFS for shortest paths.	"Fewest connections" or "fewest steps" in an unweighted graph.
3	Dynamic programming	DP pays off when subproblems overlap and have optimal substructure; memoizing solves each subproblem once.	Assuming memoization speeds up any recursion.	The same recursive call is repeated many times.
4	NP-complete problems	No known polynomial-time algorithm solves NP-complete problems exactly; practical work uses heuristics or approximations.	Believing a small-looking input can be brute-forced or that impossibility is proven.	Routing many stops, scheduling, or packing with a demand for the exact best answer.
5	Huffman compression	Huffman coding gives the most frequent symbols the shortest codes; total bits = $\sum$ frequency $\times$ code length.	Assigning short codes to rare symbols or using fixed-length totals.	A symbol frequency table and a question about bits.
6	Hash table performance	Hash lookups are $O(1)$ on average but degrade to $O(n)$ when keys collide into the same bucket.	Saying hash lookups are always $O(1)$.	A poor hash function or all keys in one bucket.
7	Self-adjusting trees	A splay tree moves each accessed node to the root, so frequently used items become fast to reach.	Treating a splay tree like a plain BST or a height-balanced AVL tree.	"Recently accessed items are accessed again."
8	Minimum spanning tree and union-find	Kruskal adds the cheapest edge that joins two different sets and skips any edge that would form a cycle; an MST on V vertices has $V - 1$ edges.	Adding every cheap edge regardless of cycles.	An edge whose endpoints are already connected.

Course scope (public description): Data Structures and Algorithms II explores the analysis and implementation of high-performance data structures and supporting algorithms, including graphs, hashing, self-adjusting data structures, set representations, and dynamic programming. The course also introduces students to NP-complete problems. The course discusses how to use Python techniques to implement software solutions for problems of memory management and data compression. This course has two prerequisites: Data Structures and Algorithms I and Discrete Math II.